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Module 13911 - Algebra: Structures and Algorithms - Lecture

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0 The Very Basics

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The key structures in algebra: groups, rings and fields.

Definition (Abelian) Group

An *abelian group* is a set G with an operation $* : G \times G \to G$ which satisfies the following $\forall a, b, c \in G$:

$$a * b = b * a \quad \text{(C)}$$
$$a * (b * c) = (a * b) * c \quad \text{(A)}$$
$$a * e = a \quad \text{(N)}$$
$$a * a^{-1} = e \quad \text{(I)}$$

$\Rightarrow$ without the commutativity (C) G would just be called a *group*.

$\rightarrow$ Group Theory (Abelian Groups are much easier than general ones)

Definition Ring

A *ring* is a set R with the operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ which satisfies the following $\forall a, b, c \in R$:

CA : $a + b = b + a, \forall a, b \in R$
AA : $a + (b + c) = (a + b) + c, \forall a, b, c \in R$
NA : There exists a unique element in R, denoted by 0 or 0_R, such that $a + 0 = a, \forall a \in R$
IA : For every $a \in R$ there exists a unique element $-a \in R$ such that $a + (-a) = 0$
AM : $a \cdot (b \cdot c) = (a \cdot b) \cdot c, \forall a, b, c \in R$
D : $a \cdot (b + c) = a \cdot b + a \cdot c, \forall a, b, c \in R$ and $(a + b) \cdot c = a \cdot c + b \cdot c, \forall a, b, c \in R$

$\rightarrow$ A ring $(R, +, \cdot)$ is called *commutative* when:

CM : $a \cdot b = b \cdot a, \forall a, b \in R$

$\rightarrow$ A ring $(R, +, \cdot)$ is called a *unitary* when:

NM : There exists a unique element in R, denoted by 1 or 1_R, such that $a \cdot 1 = a, \forall a \in R$

$\rightarrow$ A ring $(R, +, \cdot)$ is called *proper* if in that ring one has $0_R \neq 1_R$.

$\rightarrow$ Theory of commutative rings is called commutative algebra.

Definition Field

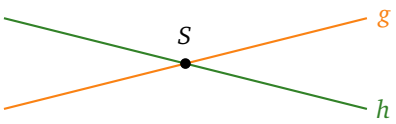
A proper, unitary, commutative ring $(R, +, \cdot)$ is called a *field* if all non-zero elements of R have a multiplicative inverse.

$\rightarrow$ Galois Theory, Theory of finite fields, algebraic closure, etc.

1 Algebra, Geometry and Algorithms

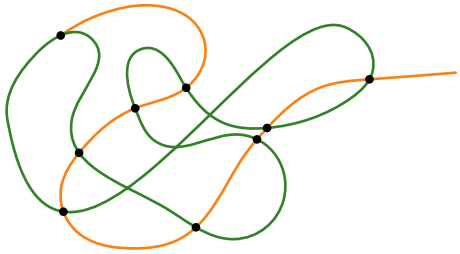
Linear Algebra

- Algebra: linear equations
- Geometry: sets defined by linear subsystems of equtions, affine subspaces, vectors, vector subspaces, ...
- Algorithms: Gaussian elimination



Non-Linear Algebra

- Algebra: polynomial equations
- Geometry: Varieties (sets defined by systems of polynomial equations)
- Algorithms: Buchberger's algorithm (a far-reading generalization of Gaussian elimination), Gröbner bases



The more Technical Layer

Notation of ideal in a ring

→ this notation is relevant for the above topics but also for construction of rings and fields (the latter is used in coding theory and cryptography)

1.1 Polynomials and the Affine Space

Variables considered per default in algebra are *formal variables (symbols)*. Formal means roughly placeholder for anything. There is no domain attached to this.

Formal variables are used in *formal expressions (symbolic expressions)*.

Note the difference to functions: Symbolic expressions are just expressions, not functions. In particular, they don't have any domain or co-domain.

In what follows, we consider n formal variables ($n \in \mathbb{Z}_{>0}$) whose default names will be $x_1, \ldots, x_n$. For $n = 1$, it is customary to use x, If $n = 2$ it's cosumary to use x and y. For $n = 3$ it's cosumary to use x, y, z.

Definition 1.1.1

Let $x = (x_1, \ldots, x_n)$ be the vector of n formal variables and $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{Z}^n_{\geq 0}$ a vector of n non-negative integers. Then the expression

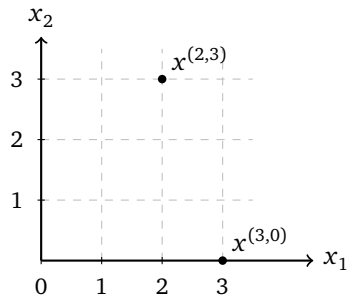
$$x^\alpha := x_1^{\alpha_1} \cdot \ldots \cdot x_n^{\alpha_n}$$

is called the *monomial for the expotent vector* α. $\alpha_1 + \ldots + \alpha_n$ is called the *(total) degree of the monomial* x^α, denoted as $|\alpha|$.

Our *coefficient domain* will be an arbitrary field which we denote by k thoughout.

Example

Given: $x^{(2,3)} = x_1^2 \cdot x_2^3$ and $x^{(3,0)} = x_1^3$



You already know some fields: $\mathbb{Q}, \mathbb{R}, \mathbb{CF}_p$ and $\mathbb{F}_{p^N}$ where p is prime, $\mathbb{Q}[\sqrt{2}]$

→ It's important to know some examples of them.

Definition 1.1.2

A *polynomial in variables* $x_1, \ldots, x_n$ with coefficients in k is a formal expression of the form

$$f = \sum_{\alpha} c_\alpha x^\alpha, \tag{1}$$

where α runs over $\mathbb{Z}^n_{\geq 0}$ and $c_\alpha \neq 0$ for any finitely many choices of k.

(You must have seen polynomials in one variable before... surprise, surprise - life is not one-dimensional!)

$c_\alpha x^\alpha$ (when $c_\alpha \neq 0$) is called a *term of degree* $|\alpha|$ of f. The *(total) degree of* f is the maximum degree of the terms in f:

$$\deg f = \max\{|\alpha| : \alpha \in \mathbb{Z}^n_{\geq 0}, c_\alpha \neq 0\}$$

The *zero polynomial* $f = 0$ has no terms and by this degree $-\infty$.
The *set of all polynomials in* $x_1, \ldots, x_n$ with coefficients in k is denoted by $k[x_1, \ldots, x_n]$.

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Note: Polynomials are formal expressions, not functions.

Equality of two polynomials is defined through comparison of coefficients.

We make $k[x_1, \ldots, x_n]$ an algebraic structure by introducing the basic operations, which are the addition and the multiplication.

Addition:

$$\sum_{\alpha} c_\alpha \cdot x^\alpha + \sum_{\alpha} d_\alpha \cdot x^\alpha := \sum_{\alpha} (c_\alpha + d_\alpha) \cdot x^\alpha$$

Multiplication:

$$\left(\sum_{\alpha} c_\alpha \cdot x^\alpha\right)\left(\sum_{\beta} d_\beta \cdot x^\beta\right) := \sum_{\alpha,\beta} c_\alpha \cdot d_\beta \cdot x^{\alpha+\beta} = \sum_{\gamma} \left(\sum_{\alpha,\beta:\alpha+\beta=\gamma} c_\alpha \cdot d_\beta\right) \cdot x^\gamma$$

Adrian's question

Adrian asked about the role of the condition $c_\alpha \neq 0$ for finitely many α. It makes the number of terms finite and allows one to define evaluation of polynomials later on. But: if one remove the finiteness assumption, what one gets is the notion of a formal power series. The set of all formal power series with coefficients in k and variables $x_1, \ldots, x_n$ is denoted as $k[[x_1, \ldots, x_n]]$. They have applications in different situations, including combinatorics (generating functions).

Remark 1.1.3

We can view k as a subset of $k[x_1, \ldots, x_n]$ and by this $k[x_1, \ldots, x_n]$ contains $1 \in k$ (The multiplecative identity).
It turns out that $(k[x_1, \ldots, x_n], +, \cdot)$ is a unitary, commutative ring:

$$f + (h + g) = (f + h) + g \qquad f \cdot (h \cdot g) = (f \cdot h) \cdot g \qquad (f + g) \cdot h = f \cdot h + g \cdot h$$
$$f + g = g + f \qquad f \cdot g = g \cdot f$$
$$f + 0 = f \qquad f \cdot 1 = f$$
$$f + (-f) = 0$$

This holds for all $f, g, h \in k[x_1, \ldots, x_n]$. The $-f$ is the unique polynomial whose addition to f gives 0.

Remark 1.1.4

There are three basic choices of n:

$n = 1$	x_1	x
$n = 2$	x_1, x_2	x, y
$n = 3$	x_1, x_2, x_3	x, y, z

Later, a connection to geometry will be established. These three basic cases will get connected to line resp. plane and resp. 3-dim. space.

Definition 1.1.5

In algebraic geometry, the space k^n of all n-tuples with the coefficients in k is called the *n-dimentional affine space over the field k*.
From linear algebra we know, that k^n is a k-vector space.

Fun Fact: It's by far not the only space one can consider. Another very common and important space is the projective space. Actually, in the Elliptic Curve Cryptography (ECC), the elliptic curve is considered in the projective space.

Definition 1.1.6

We can evaluate $f = \sum_\alpha c_\alpha \cdot x^\alpha \in k[x_1, \ldots, x_n]$ at a point $(a1, \ldots, a_n) \in k^n$ of the affine space obtaining the value.

$$f(a_1, \ldots, a_n) = \sum_{\alpha=(\alpha_1,\ldots,\alpha_n)} c_\alpha a_1^{\alpha_1} \cdots a_n^{\alpha_n}$$

This gives rise to to a function $k^n \to k$. This function is usually denoted again by f, even though $f \in k[x_1, \ldots, x_n]$ and $f : k^n \to k$ is not the same thing. In math, such a situation is called abuse of notation.

I know it's not quite right, but I know what I'm doing

Functions arising this way are called *polynomial functions*.

Remark 1.1.7

In some fields, for example, in finite fields k, different polynomials in $k[x_1, \ldots, x_n]$ may define the same polynomial function.

Example

Consider the finite field $\mathbb{F}_2 = \{0,1\}$ with

$+$	0	1
0	0	1
1	1	0

and

$\cdot$	0	1
0	0	0
1	0	1

$f = x^2 + x \in \mathbb{F}_2[x]$
$g = x^3 + x \in \mathbb{F}_2[x]$

$f : \mathbb{F}_2 \to \mathbb{F}_2$

$a \in \mathbb{F}_2$	$f(a) \in \mathbb{F}_2$
0	0
0	0

$g : \mathbb{F}_2 \to \mathbb{F}_2$

$a \in \mathbb{F}_2$	$g(a) \in \mathbb{F}_2$
0	0
0	0

So, both functions are the same, but the polynomials are different

Fun Fact: Do we need $\mathbb{F}_2[x]$ for anything?
→ Yes! Advanced Encryption Standard (AES) used the field $\mathbb{F}_{256}$, whose construction requires taking a polynomial from $\mathbb{F}_2[x]$.

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Note: Test today → some will need to catch up
→ Check-In Date for the catch-up cohort: 9th of November, you show your progress on the form of the written solutions of the exercises from a textbook (linear algebra over arbitrary fields, with proofs).

Entry Level Exercise

1. Formulate mathematical induction (with words and/or formulas).
2. Define the notion of a group. Give an example of an abelian and a non-abelian group.
3. Define the notion of a ring. Give an example of a commutative and a non-commutative ring.
4. Define the notion of a field. Give an example of an infinite and finite field.
5. Let k be a infinite field and consider a univariate polynom $f \in k[x]$.
   Proof: $f = 0 \iff f(a) = 0, \forall a \in k$.

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Theorem 1.1.8

Let the field k be infinite. Then $f \in k[x_1, \ldots, x_n]$ is a zero polynomial (all coefficients of f are zero) if and only if $f : k^n \to k$ is a zero function.

Proof: The implikation ($\Rightarrow$) is trivial because $f = \sum_\alpha c_\alpha x^\alpha$ gets evaluated to zero on k^n when all of the c_α's are zero. So, the actual work is to show the converse ($\Leftarrow$). For the converse, we are going to use the mathematical induction on n.

We consider the case $n = 1$. as a basis of the induction. We consider a polynomial f in one variable x, which is <u>not</u> a zero polynomial and show that $f(a) \neq 0$ for some $a \in k$. We can write f as $f = \sum_{i=1}^{d} c_i x^i$ where $d \in \mathbb{N}_0$ and $c_d \neq 0$.

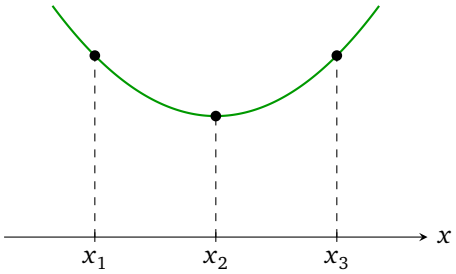
> **Version 1 of the argument:** Pick a zero of f, as long as there is one, let's call it $a_1 \in k$ and divide f by $(x - a_1)$ (in the sense of long division). It get's divided with the 0 remainder. So, we have $f = (x = a_1) \cdot g$ where g has degree $d - 1$ (the degree drops). If g is divisible by $(x - a_1)$, keep dividing. Once this process of dividing by $(x - a_1)$ is finished, we have $f = (x - a_1)^{p_1} \cdot h(x)$, where the degreee of h is smaller than the degree of g and a_1 is not a zero of h. If there is another zero, say $a_2 \in k$, do the same obtaining $f = (x - a1)^{p_1} \cdot (x - a_2)^{p_2} \cdot u(x)$, where a_1 and a_2 are both not zero of $u(x)$. Iterating this way, we end up at the representation $f = (x - a_1)^{p_1} \cdot (x - a_2)^{p_2} \cdot \cdots \cdot (x - a_m)^{p_m} \cdot w(x)$ where $a_1, \ldots, a_m \in k$, $degf = p_1 + \ldots + p_m + deg(w)$ and w has no zero in k. This shows that a non-zero polynomial of degree $d \in \mathbb{N}_0$ with at most $d \in \mathbb{N}_0$ zeros, where d is finite, but our field k is infinite. So, $f : k \to k$ is only zero on $\{a_1, \ldots, a_m\}$ and so it's not a zero function.

Theorem 1.1.8 (Continued)

Version 2 of the argument for $n = 2$: Let's use linear algebra and polynomial interpolation. Let's consider all polynomial of degree at most $d \in \mathbb{N}$, where d is fixed and arbitrary. Denote the set of these polynomials by $k[x]_d = \{\sum_{i=0}^{d} c_i x^i : c_i \in k \text{for every} i\}$. This is a k-vector space of dimension of $d+1$. We fix $d+1$ dstinct values $a_1, \ldots, a_{d+1}$ in k (our field is infinite, so no matter how large our d is, we always have such a choice) and consider the linear map that evaluates $f \in k[x]_d$ at these values.

$$ev(f) = (f(a_1), \ldots, f(a_{d+1}))$$

This is the map $ev : \underbrace{k[x]_d}_{\text{dimension } d+1} \to \underbrace{k^{d+1}}_{\text{dimension } d+1}$.



We're going to show that this map is surjective (in the language of the numerical mathematics: the polynomial interpolation task is solvable).

Let's fix an arbitrary $y = (y_1, \ldots, y_{d+1}) \in k^{d+1}$ and determine an $f \in k[x]_{d+1}$ satisfying $f(a_i) = y_i$ for all $i \in \{1, \ldots, d+1\}$.
Introduce

$$l_i(x) = \prod_{j=1\ldots,d+1: j \neq i} \frac{x - a_j}{a_i - a_j} \in k[x]_d$$

which satisfies

$$l_i(a_j) = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

Now, it is clear that $f = y_1 \cdot l_1 + \ldots y_{d+1} \cdot l_{d+1}$ satisfies $f(a_i) = y_i$ for all $i = 1, \ldots, d+1$. We have shown that $ev : k[x]_d \to k^{d+1}$ is surjective. So, from linear algebra, we know that f is also injective so that only the zero polynomial is mapped to $(0, \ldots, 0)$ by ev.

If you like more details on that point: by the rank-nullity theorem,

$$\underbrace{\dim \operatorname{im}(ev)}_{\text{dimension} d+1} + \underbrace{\dim \ker(ev)}_{\text{dimension} d+1} = \dim k[x]_d,$$

so that $\dim \ker(ev) = 0$.

We have done the case $n = 1$ and we continue with the inductive argument. Assume that the assertion is already verified for polynomials in $n-1$ variables for some $n \geq 2$. Based on this, we verify the assertion for polynomials in n variables.

We can write a polynomial $f \in k[x_1, \ldots, x_n]$ as an element of $k[x_1, \ldots, x_{n-1}][x_n]$, so we write $f = \sum_{i=1}^{d} g_i(x_1, \ldots, x_{n-1}) \cdot x_n^i$ with $g_1, \ldots, g_d k[x_1, \ldots, x_{n-1}]$. We fix arbitrary values for $x_1, \ldots, x_{n-1}$, which we denote by $a_1, \ldots, a_{n-1} \in k$. Fixing these values, we get $f(a_1, \ldots, a_{n-1}, x_n) = \sum_{i=1}^{d} g_i(a_1, \ldots, a_{n-1}) \cdot x_n^i \in k[x_n]$. By the $n = 1$ case of our argument, whenever $f(a_1, \ldots, a_{n-1}, x_n)$ defines the zero function $k \to k$, it is a zero polynomial. Hence, its coefficients $g_i(a_1, \ldots, a_{n-1})$ are a;; zero. But since $(a_1, \ldots, a_{n-1})$ are arbitrary values from k, the induction assumption implies that $g_i(x_1, \ldots, x_{n-1})$ is also a zero polynomial. Consequently, f is a zero polynomial.

□